

# EFFECT OF STRENGTH AND HIGH-INTENSITY TRAINING ON JUMPING, SPRINTING, AND INTERMITTENT ENDURANCE PERFORMANCE IN PREPUBERTAL SOCCER PLAYERS

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## ABSTRACT

Ferrete, C, Requena, B, Suarez-Arrones, L, and Sáez de Villarreal, E. Effect of strength and high-intensity training on jumping, sprinting, and intermittent endurance performance in prepubertal soccer players. *J Strength Cond Res* 28(2): 413–422, 2014—The purpose of this study was to examine the effects of a 26-week on-field combined strength and high-intensity training on the physical performance capacity among prepubertal soccer players who were undertaking a competitive phase of training. Twenty-four prepubertal soccer players between the age of 8 and 9 years were randomly assigned to 2 groups: a control (C;  $n = 13$ ) and an experimental group (S;  $n = 11$ ). Both groups performed an identical soccer-training program, whereas the S group also performed combined strength and high-intensity training before the soccer-specific training. The 15-m sprint time (seconds), countermovement jump (CMJ) displacement, Yo-Yo intermittent endurance test (Yo-Yo IE), and Sit and Reach flexibility were each measured before (baseline) and after 9 (T2), 18 (T3), and 26 weeks (posttest) of training. There were no significant differences between the groups in any of the variables tested at baseline. After 26 weeks, significant improvements were found in the CMJ (6.72%; effect size [ES] = 0.37), Yo-Yo IE (49.57%, ES = 1.39), and Flexibility (7.26%; ES = 0.37) variables for the S group. Conversely, significant decreases were noted for the CMJ (−10.82%; ES = 0.61) and flexibility (−13.09%; ES = 0.94) variables in the C group. A significant negative correlation was found between 15-m sprint time and CMJ ( $r = -0.77$ ) and Yo-Yo IE ( $r = -0.77$ ) in the S group. Specific combined strength and high-intensity training in prepubertal soccer players for 26 weeks produced a positive effect on performance qualities highly specific to soccer. Therefore, we propose modifications to current training methodology for prepubertal soccer players to

include strength and high-intensity training for athlete preparation in this sport.

**KEY WORDS** sprint, Yo-Yo test, countermovement jump, flexibility, football

## INTRODUCTION

Soccer is one of the most popular sports in the world with the age of participants spanning from childhood to old age and is marked by various intensities of physical activity. As a whole, these various movement intensities are performed in an intermittent fashion, and during game situations, there are moments where relatively low-intensity movements are interspersed with maximal bouts of high-intensity exercise (36). Specifically, these high-intensity movements (i.e., sprinting, jumping, or cutting) and low-intensity movements (jogging or standing) occur in varying lengths depending upon an array of factors that include the athlete's playing position, skill level of the athlete, style of play, and tactical strategies employed by the team (24). Careful inspection of a match reveals that high-speed sprinting only contributes approximately 3% to the total distance covered in children's games (4), and the most crucial moments of the game such as winning ball possession, scoring, or conceding goals depend on the ability of the athlete to perform these high-speed movements (35). It is generally accepted that high-intensity actions such as sprinting or vertical jumping are integral elements for success in soccer and therefore need to be trained as part of a periodized youth training program (23).

Recently, numerous studies have focused on determining which training intervention maximize improvements in sprint and jumping performance in both adult and young soccer players (7,8,30). The vast majority of the scientific studies looking at training interventions designed to improve these attributes have proposed in-season short-term training (e.g., 4–8 weeks) programs that employ several weekly training interventions (7,8,12,16,18,30). Ultimately, these training interventions target the development of sprint or jumping performance and attempt to integrate with other training

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factors to increase recovery from matches and soccer-specific training sessions. For example, Mujika et al. (33) showed a significant improvement in 15-m sprint time after a 7-week period in which the specific soccer training was supplemented with 1 session per week of complex training (alternating heavy-light resistance with soccer-specific drills) in elite youth soccer players. One important aspect of the results presented by Mujika et al. (33) is that the use of 1 training session dedicated to supplemental training was able to improve performance. This finding may be of particular interest to soccer coaches because 1 training session of this type could potentially be added to the competitive soccer schedule, regardless of player age or how busy the competitive schedules to maintain fitness during the competitive season (36).

In young soccer players, several training interventions have proven to be effective for improving vertical jump and sprint performance. Mainly, these interventions have included traditional resistance training methods with submaximal resistances (i.e., 30–60% 1 repetition maximum) (7,8,25), complex training methods (33,44), and plyometric training interventions (12,30). For example, Christou et al. (8) after 16 weeks of traditional resistance training that was performed twice a week in addition to soccer training (5 times per week) observed greater increases in upper- and lower-body strength, and vertical jump performance (i.e., Squat Jump [SJ] was increased 31%) compared with soccer training alone (i.e., SJ was increased 9.8%). Similarly, Meylan and Malatesta (30) observed that 8 weeks of low-intensity plyometric training (2 training interventions per week) implemented in conjunction with the soccer-specific training program of adolescent soccer players resulted in significantly improved 10-m sprint times (0.04 seconds, –2.1%) and jump heights (countermovement jump [CMJ], 2.6 cm, 7.9%). All these training strategies suggest that the inclusion of a more intense and non-soccer-specific training stimulus, which is integrated into the normal soccer-training program, has the potential to induce improvements in jumping and sprinting performances that are greater than maturation or soccer-specific training performance gains.

Additionally, it is widely accepted that the capacity to perform repeated explosive bouts is an important determinant of player performance (24), and it is also associated with a high aerobic power ( $\dot{V}O_{2\max}$ ) (18). Because of the relationship between  $\dot{V}O_{2\max}$  and the ability to engage in repeated explosive bouts, it is often believed that the implementation of a soccer-training program at an early age that contains a strength-based training intervention may negatively impact aerobic endurance. The majority of the studies discussed previously have only examined the training effects of strength training on explosive movements such as sprint and jump, and have not examined the specific effects of these training interventions on aerobic endurance. Similarly, to the authors' knowledge, there is little information in the scientific literature about the effects of these proposed training interventions on joint flexibility, especially hip flexibility.

To date, only 1 study (8) has measured hip flexibility after the training and showed a reduction of 8% in the sit and reach test in comparison with soccer training alone. Typically, a reduced range of motion in hip joint has been associated with an increased hamstrings injury risk in soccer players (19) and the inclusion of a strength program that compromises the hips range of motion may be problematic.

Nevertheless, to the best of our knowledge, no study has analyzed if similar effects occur in prepubertal soccer players (<10 years). In addition, only short-term training interventions (mostly between 4 and 8 weeks) have been discussed in the scientific literature. To the authors' knowledge, no study has examined the effects of a long-term training intervention (>24 weeks) in prepubertal soccer players. In light of the aforementioned considerations, the aim of this study was to examine the effects of an in-season, low-impact (low-volume) strength, and high-intensity (sprinting, jumping and cutting) training program on physical performance among prepubertal soccer players. It was hypothesized that the combination of soccer drills and the proposed strength and high-intensity training program during a 26-weeks' period would enhance players' vertical jump and sprint performance to a greater extent than soccer drills alone without modifying endurance fitness and hip joint flexibility.

## METHODS

### Experimental Approach to the Problem

This study was designed to determine the effects of a strength and high-intensity training program on the prepubescent soccer players vertical jump performance, sprint speed, endurance, and hip flexibility. The training program was integrated into the players' normal soccer training during a 26-week period that was mainly characterized by low-volume and high-intensity (strength, sprinting, jumping, and cutting) sessions, which was considered a moderate-impact training program. To determine training effects, the following tests were selected: (a) 15-m sprint time, (b) CMJ height, (c) Yo-Yo intermittent endurance test (Yo-Yo IE), and (d) Sit and Reach flexibility test. All the tests were executed before the initiation of the training period and were started at baseline (baseline), after 9 weeks (T2), 18 weeks (T3), and at the end of the training period (26 weeks) (posttest). The initial tests were completed in 2 days (Monday and Wednesday) as part of a regularly scheduled testing program. After the initial measurements, the subjects were randomly assigned to 2 groups: control (C;  $n = 13$ ) that only performed the soccer-training program, and an experimental group (S;  $n = 11$ ) that performed the same soccer-training program as the C group plus a specifically designed strength and high-intensity training program.

### Subjects

This study involved a group of 24 young soccer players from the Real Betis Balompié Academy, in Spain. All the recruited subjects were between the ages of 8 and 9 years (Table 1) and did not have any background involving regular strength

**TABLE 1.** Characteristics of the groups (mean  $\pm$  SD).

| Group          | Age (y)         | Height (cm)     |                  | Body weight (kg) |                 | % Body fat     |                | Soccer exp (mos) |
|----------------|-----------------|-----------------|------------------|------------------|-----------------|----------------|----------------|------------------|
|                |                 | Pre             | Post             | Pre              | Post            | Pre            | Post           |                  |
| C ( $n = 13$ ) | 8.26 $\pm$ 0.33 | 131.9 $\pm$ 6.6 | 136.6 $\pm$ 6.1* | 29.3 $\pm$ 4.6   | 30.9 $\pm$ 3.9* | 12.8 $\pm$ 1.9 | 12.2 $\pm$ 3.9 | 18               |
| S ( $n = 11$ ) | 9.32 $\pm$ 0.25 | 135.6 $\pm$ 6.2 | 145.1 $\pm$ 6.6* | 34.1 $\pm$ 4.2   | 35.9 $\pm$ 4.3* | 13.2 $\pm$ 0.5 | 12.8 $\pm$ 3.1 | 24               |

\*Significant difference between pretraining and posttraining values ( $p \leq 0.05$ ).

and high-intensity training or competitive sports that involved any kind of strength or high-intensity exercises before the initiation of this study. Exclusion criteria for this study included subjects with potential medical problems or a history of ankle, knee, or back pathology in the 3 months before the initiation of the study. Additionally, the subjects with medical, orthopedic problems that compromised their participation or performance in this study, any lower extremity reconstructive surgery in the past 2 years, or unresolved musculoskeletal disorders were excluded from participation. All the subjects and their legal guardians were carefully informed about the experiment procedures and about the potential risk and benefits associated with participation in the study. All the subjects and guardians read and signed an informed consent document before being included in the study. All the procedures were approved by an Institutional Ethics Review Committee of the University in accordance with the current national and international laws and regulations governing the use of human subjects (Declaration of Helsinki II).

#### Testing Procedures

The players were familiarized with all the tests and procedures used to evaluate force and power production before the initiation of testing. In addition to familiarization sessions that were performed a few days before testing, all the subjects participated in testing sessions using similar protocols while training with their Academy programs. All tests to determine the vertical jump, sprint, endurance, and hip-flexibility capacity were carried out every 9 weeks (baseline, T2, T3, and posttest), throughout the 26 weeks that encompassed this study. The performance tests were completed in 2 days: On day 1: anthropometric measurements, height of CMJ (centimeters) and 15-m sprint time tests were completed. On day 2, the flexibility test (sit and reach) and intermittent endurance test (Yo-Yo IE) were completed. Before the tests and after completing the anthropometric measurements, the subjects carried out a standardized warm-up consisting of 5 minutes of submaximal running at  $9 \text{ km} \cdot \text{h}^{-1}$  followed by dynamic light stretching. The subjects then performed 2 minutes of specific soccer warm-up (changes of directions, sprints, jumps, and heading) and 4 minutes of stretching (first static and then dynamic). Additionally, sufficient rest was carefully allowed between all tests.

**Anthropometry.** Before the physical tests, body height, body weight (Seca-balance, Seca 222, New York, NY, USA), and body fat percentage of the subjects were determined. The fat percentage was calculated by means of measurements of skinfold thickness using a Harpenden skinfold calliper (ASSIST Creative Resources Ltd., London, United Kingdom). The 7-site Jackson-Pollock formula (41), validated for use with athletes, was used in this study to estimate body density. Body fat percentage was then calculated using the appropriate formula recommended by American College of Sports Medicine (2), based on age and ethnicity. Two experienced testers assessed the anthropometric measurements throughout the entire study.

**Countermovement Jump.** A CMJ was used to assess explosive strength of the lower extremity muscles. The CMJ test was performed using an electronic contact platform (Ergo Jump Plus Bosco System, Muscle LabV718, Langesund, Norway). During the CMJ, the subjects were instructed to place their hands on their hips while performing a downward movement followed by a maximal effort vertical jump. All the subjects were instructed to land in an upright position and to bend the knees after landing. Five trials were executed with a pause of 5 seconds between jumps. The 2 extreme values of the 5 trials were eliminated (best and worst), and the average of the 3 central values was used for the subsequent statistical analysis. The intraclass correlation coefficient (ICC) for this test was 0.96.

**The 15-m Sprint Time.** Sprint times were recorded for a 15-m distance that was conducted outdoors with suitable weather conditions (sunny and standard conditions of wind) on the soccer pitch (artificial grass). For all sprint tests, the subjects started the test using a crouch start and commenced sprinting with a random sonorous sound. Infrared beams were positioned at the sprint distance to be measured with photoelectric cell (Muscle LabV718). After a thorough warm-up, the subjects were given 2 practice trials performed at half speed to familiarize them with the timing device. Two trials were completed, and the best performance trial was used for the subsequent statistical analysis. Three minutes of rest was permitted between 15-m trials. The reliability of the 15-sprint test was considered to be high as indicated by an ICC = 0.94.

**TABLE 2.** Strength training program for the S group.\*

|                             | 1        |          | 2        |          | 3        |          | 4        |          | 5        |          | 6        |          | 7        |          | 8        |          |       |
|-----------------------------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|-------|
| Weeks                       |          |          |          |          |          |          |          |          |          |          |          |          |          |          |          |          |       |
| Exercises/<br>sessions      | S1       | S2       | S3       | S4       | S5       | S6       | S7       | S8       | S9       | S10      | S11      | S12      | S13      | S14      | S15      | S16      | 9     |
| 1/4 Squat                   | 2 ×<br>8 | 3 ×<br>8 | 3 ×<br>8 | 3 ×<br>6 | 3 ×<br>8 | 3 ×<br>8 | 2 ×<br>6 | 3 ×<br>6 | 3 ×<br>6 | 3 ×<br>4 | 3 ×<br>4 | 3 ×<br>4 | 2 ×<br>6 | 3 ×<br>6 | 3 ×<br>6 | 3 ×<br>6 | Tests |
| Vertical jump               | 3 ×      |          | 3 ×      |          | 3 ×      |          | 3 ×      |          | 3 ×      |          | 3 ×      |          | 3 ×      |          | 3 ×      |          |       |
| (3-kg weight)               | 6        |          | 6        |          | 6        |          | 6        |          | 6        |          | 6        |          | 6        |          | 6        |          |       |
| Deep squat                  | 3 ×      |          | 3 ×      |          | 3 ×      |          | 3 ×      |          | 3 ×      |          | 3 ×      |          | 3 ×      |          | 3 ×      |          |       |
|                             | 6        |          | 6        |          | 6        |          | 6        |          | 6        |          | 6        |          | 6        |          | 6        |          |       |
| Weight                      | 3 ×      |          | 3 ×      |          | 4 ×      |          | 4 ×      |          | 5 ×      |          | 3 ×      |          | 4 ×      |          | 4 ×      |          |       |
| displacement                | 10"      |          | 10"      |          | 10"      |          | 10"      |          | 10"      |          | 10"      |          | 10"      |          | 10"      |          |       |
| Obstacle jump               |          | 3 ×      |          | 3 ×      | 3 ×      |          | 3 ×      |          |          | 3 ×      |          | 3 ×      |          | 3 ×      |          | 3 ×      |       |
|                             |          | 5        |          | 5        | 5        |          | 5        |          |          | 5        |          | 5        |          | 5        |          | 5        |       |
| Sprint                      |          | 3 ×      |          | 4 ×      | 3 ×      |          | 4 ×      |          |          | 4 ×      |          | 3 ×      |          | 4 ×      |          | 4 ×      |       |
|                             |          | 20       |          | 20       | 20       |          | 20       |          |          | 20       |          | 20       |          | 20       |          | 20       |       |
|                             | 10       |          | 11       |          | 12       |          | 13       |          | 14       |          | 15       |          | 16       |          | 17       |          |       |
| Weeks                       |          |          |          |          |          |          |          |          |          |          |          |          |          |          |          |          |       |
| Exercises/<br>sessions      | S17      | S18      | S19      | S20      | S21      | S22      | S23      | S24      | S25      | S26      | S27      | S28      | S29      | S30      | S31      | S32      | 18    |
| 1/4 Squat                   | 3 ×<br>6 | 3 ×<br>6 | 3 ×<br>8 | 3 ×<br>6 | 3 ×<br>8 | 3 ×<br>8 | 3 ×<br>6 | 3 ×<br>6 | 3 ×<br>6 | 3 ×<br>4 | 3 ×<br>4 | 3 ×<br>4 | 3 ×<br>4 | 3 ×<br>4 | 3 ×<br>6 | 3 ×<br>6 | Tests |
| Vertical jump               | 3 ×      |          | 3 ×      |          | 3 ×      |          | 3 ×      |          | 3 ×      |          | 3 ×      |          | 3 ×      |          | 3 ×      |          |       |
| (3-kg weight)               | 4        |          | 4        |          | 4        |          | 4        |          | 6        |          | 6        |          | 6        |          | 5        |          |       |
| Deep squat                  | 3 ×      |          | 3 ×      |          | 3 ×      |          | 3 ×      |          | 3 ×      |          | 3 ×      |          | 3 ×      |          | 3 ×      |          |       |
|                             | 6        |          | 6        |          | 6        |          | 6        |          | 6        |          | 6        |          | 6        |          | 6        |          |       |
| Weight                      | 4 ×      |          | 4 ×      |          | 4 ×      |          | 4 ×      |          | 5 ×      |          | 3 ×      |          | 3 ×      |          | 3 ×      |          |       |
| displacement                | 10"      |          | 10"      |          | 10"      |          | 10"      |          | 10"      |          | 10"      |          | 10"      |          | 10"      |          |       |
| Obstacle jump               |          | 3 ×      |          | 3 ×      |          | 3 ×      |          | 3 ×      |          | 3 ×      |          | 3 ×      |          | 3 ×      |          | 3 ×      |       |
|                             |          | 5        |          | 5        |          | 5        |          | 5        |          | 5        |          | 5        |          | 5        |          | 5        |       |
| Sprint                      |          | 4 ×      |          | 4 ×      |          | 3 ×      |          | 4 ×      |          | 4 ×      |          | 3 ×      |          | 3 ×      |          | 4 ×      |       |
|                             |          | 20       |          | 20       |          | 20       |          | 20       |          | 20       |          | 20       |          | 20       |          | 20       |       |
|                             | 19       |          | 20       |          | 21       |          | 22       |          | 23       |          | 24       |          | 25       |          | 26       |          |       |
| Weeks                       |          |          |          |          |          |          |          |          |          |          |          |          |          |          |          |          |       |
| Exercises/<br>sessions      | S33      | S34      | S35      | S36      | S37      | S38      | S39      | S40      | S41      | S42      | S43      | S44      | S45      | S46      | S47      | S48      | 27    |
| 1/4 Squat                   | 4 ×<br>6 | 4 ×<br>6 | 4 ×<br>8 | 4 ×<br>6 | 4 ×<br>8 | 4 ×<br>8 | 5 ×<br>6 | 5 ×<br>6 | 5 ×<br>6 | 6 ×<br>4 | 4 ×<br>4 | 3 ×<br>4 | 4 ×<br>4 | 3 ×<br>4 | 4 ×<br>6 | 4 ×<br>4 | Tests |
| Vertical jump (3-kg weight) | 4 ×      |          | 4 ×      |          | 4 ×      |          | 5 ×      |          | 3 ×      |          | 2 ×      |          | 2 ×      |          | 4 ×      |          |       |
| Deep squat                  | 4 ×      |          | 4 ×      |          | 4 ×      |          | 5 ×      |          | 5 ×      |          | 3 ×      |          | 3 ×      |          | 4 ×      |          |       |
|                             | 6        |          | 6        |          | 6        |          | 6        |          | 6        |          | 6        |          | 6        |          | 6        |          |       |
| Weight                      | 5 ×      |          | 5 ×      |          | 5 ×      |          | 6 ×      |          | 5 ×      |          | 3 ×      |          | 3 ×      |          | 4 ×      |          |       |
| displacement (in.)          | 10       |          | 10       |          | 10       |          | 10       |          | 10       |          | 10       |          | 10       |          | 10       |          |       |
| Obstacle jump               |          | 4 ×      |          | 4 ×      |          | 4 ×      |          | 5 ×      |          | 5 ×      |          | 3 ×      |          | 3 ×      |          | 4 ×      |       |
|                             |          | 5        |          | 5        |          | 5        |          | 5        |          | 5        |          | 5        |          | 5        |          | 5        |       |
| Sprint                      |          | 4 ×      |          | 4 ×      |          | 4 ×      |          | 4 ×      |          | 5 ×      |          | 3 ×      |          | 3 ×      |          | 4 ×      |       |
|                             |          | 20       |          | 20       |          | 20       |          | 20       |          | 20       |          | 20       |          | 20       |          | 20       |       |

\*S = sessions; 1/4 squat = the degree of knee flexion was over 120°; vertical jump (3-kg weight) = rebounds jumps using a weight ball of 3 kg; deep squat = the degree of knee flexion was >30°; weight displacement = maximal speed with partner resistance (a set always was of 10 seconds); obstacle jump = rebounds jumps over platform of 5, 10, and 15 cm and lateral jumps over hurdles of 10 cm; sprint = maximal speed on 20 m.

**TABLE 3.** Sprint 15 m (seconds), CMJ (centimeters), Yo-Yo IE (meters), and sit and reach (centimeters) test performance of the S and C groups before and after 26 weeks in-season training.\*†

| S (N = 11)         |                  |             |             |               |             |      |
|--------------------|------------------|-------------|-------------|---------------|-------------|------|
|                    | Baseline         | T2          | T3          | Post          | % Of change | ES   |
| Sprint 15 m (s)    | 2.7 ± 0.1        | 2.8 ± 0.1   | 2.9 ± 0.1   | 2.8 ± 0.1     | 3.70        | 1    |
| CMJ (cm)           | 22.3 ± 2.7‡§¶    | 23.7 ± 3.5  | 23.7 ± 3.4  | 23.8 ± 4.3    | 6.72        | 0.37 |
| Yo-Yo IE (m)       | 476 ± 169.1‡¶#   | 580 ± 193.4 | 704 ± 156.8 | 712 ± 228.4   | 49.57       | 1.39 |
| Sit and reach (cm) | 23.4 ± 4.5§¶#**  | 22.5 ± 4.5  | 25.3 ± 3.7  | 25.1 ± 5.5    | 7.26        | 0.37 |
| C (N = 13)         |                  |             |             |               |             |      |
|                    | Baseline         | T2          | T3          | Post          | % Of change | ES   |
| Sprint 15 m (s)    | 2.9 ± 0.1        | 2.9 ± 0.1   | 3.0 ± 0.1   | 2.9 ± 0.1     | 1.37        | 0.40 |
| CMJ (cm)           | 20.2 ± 3.4¶      | 20.3 ± 3.2  | 20.2 ± 2.7  | 18.0 ± 3.6    | -10.82      | 0.61 |
| Yo-Yo IE (m)       | 540.3 ± 192.5‡§¶ | 584 ± 236.4 | 640 ± 226.9 | 646.6 ± 229.9 | 19.67       | 0.55 |
| Sit and reach (cm) | 25.2 ± 3.5¶††    | 23.8 ± 2.9  | 24.2 ± 4.2  | 21.9 ± 4.7    | -13.09      | 0.94 |

\*Baseline = baseline values; T2 = after 9 weeks; T3 = after 18 weeks; POST = after 26 weeks; % of change = percentage of change between baseline-post; ES = effect size; CMJ = countermovement jump; Yo-Yo IE = Yo-Yo intermittent endurance test; S = experimental; C = control.

†Values are reported as mean ± SD.

‡Significant difference between baseline and T2 values ( $p < 0.05$ ).

§Significant difference between baseline and T3 values.

¶Significant difference between baseline and postvalues.

||Significant differences between experimental and control groups ( $p < 0.05$ ).

#Significant difference between T2 and T3 values.

\*\*Significant difference between T2 and postvalues.

††Significant difference between T3 and posttest values.

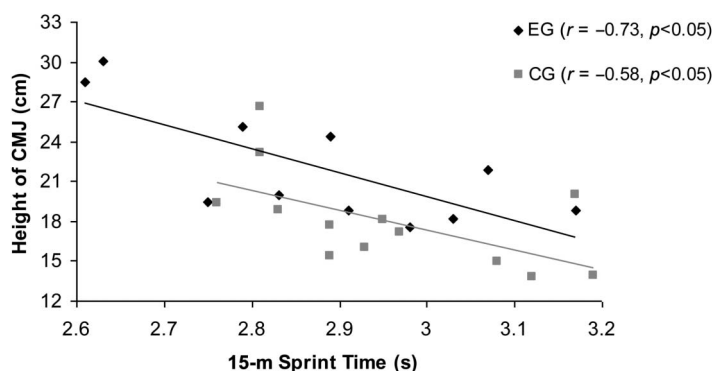
**Yo-Yo Intermittent Endurance Run: Level One.** The Yo-Yo IE test was performed according to the procedures suggested by Castagna et al. (5) and Krstrup et al. (28). Because soccer includes high-intensity, intermittent bouts of exercise, which stresses the anaerobic glycolysis metabolic pathway, the Yo-Yo IE is considered to closely match movement patterns seen in a soccer match. In this study, the Yo-Yo IE test

started in stage 8, at  $11.08 \text{ km} \cdot \text{h}^{-1}$  in standard 20-m intervals. The total distance covered in the shuttles was recorded for analysis. The ICC was 0.90.

**Flexibility (Sit and Reach) Test.** This test was used to assess the progress in the lumbar and hamstring flexibility. The sit and reach test was performed according to the procedure suggested by Wells et al. (45). Two trials were completed, with a pause of 30 seconds between trials, choosing the best performance for the subsequent statistical analysis. The reliability of this measure was  $\text{ICC} = 0.94$ .

#### Training Procedures

Soccer training took place  $3 \text{ d} \cdot \text{wk}^{-1}$  (Monday-Wednesday-Friday). Although C only underwent soccer training, the S supplemented the soccer training with a proposed strength and high-intensity training program. This program consisted of exercises



**Figure 1.** Correlation between counter movement jump (CMJ) (centimeters) and 15-m sprint time (seconds) in the S and C groups at posttest.

performed at maximal voluntary intensity using player's body weight (or body weight plus light resistances) as external resistance. The strength and high-intensity training program took place 2 d·wk<sup>-1</sup> during 26 weeks of treatment. Each session lasted 30 minutes and consisted of the following components: 10 minutes of standard warm-up (5 minutes of submaximal running at 6 km·min<sup>-1</sup> and several displacements, stretching exercises for 5 minutes, and 2 submaximal exercises of jump), 15 minutes of strength and high-intensity work, and 5 minutes of stretching exercises. The strength and high-intensity exercises consisted of 1/4 squat, deep jumps, CMJ with weight, and sprint exercises. The resting period between each series was 1 minute. No strength and high-intensity training was performed by the C group. The training was performed on the artificial grass (the same as competition), with the subjects using appropriated soccer equipped (boots and clothes). All the training sessions were fully supervised by a certified strength and conditioning specialist to ensure that the proper technique was performed. All the subjects were carefully instructed before the treatment and received a practical demonstration and performed familiarization trials with the exercises. The treatment was performed before the start of the regular soccer-training session. The subjects were instructed to avoid any strenuous physical activity, in addition to the programmed training intervention, for the duration of the experiment. Additionally, the subjects were encouraged to maintain their normal hydration levels, sleep, and dietary habits for the duration of the study. The training program followed by the S is outlined in Table 2.

#### Statistical Analyses

Descriptive statistics (mean  $\pm$  SD) for the different variables were calculated. The ICC was used to determine the reliability of the measurements. A 2-way mixed analysis of variance was used on each continuous dependent variable. The inde-

pendent variables included 1 between-subjects factor, the strength and high-intensity training intervention, with 2 levels (S and C), and 1 within-subject factor, time, with 4 levels (baseline, T2, T3, and posttest). Any significant differences found by the analysis of variance were followed by Bonferroni's post hoc analysis. A Pearson's correlation was used to examine the relationships between variables. Effect sizes (ESs) were also calculated using Cohen's  $d$  ( $\frac{[\text{posttest mean} - \text{pretest mean}]}{SD}$ ). Statistical significance was accepted at an alpha level of  $p \leq 0.05$ .

## RESULTS

### Anthropometric Characteristics

No significant differences in the anthropometric variables measured (body weight, height, and % body fat) were observed in the pretest between the S and C. Both groups increased significantly in body height and weight from the baseline to posttest with no change in percentage of body fat (Table 1). In addition, at the posttest, the S grew significantly more than the C (enhancement of 9.5 vs. 4.7 cm, respectively), whereas no differences were observed between groups at body weight.

### The 15-m Sprint Time

The 15-m Sprint Time (seconds) increased in the S (3.70%; ES = 1) and C group (1.37%; ES = 0.4). No significant differences ( $p \leq 0.05$ ) were observed after training in the magnitude of the increase between the S and C groups (Table 3).

### Height in Countermovement Jump

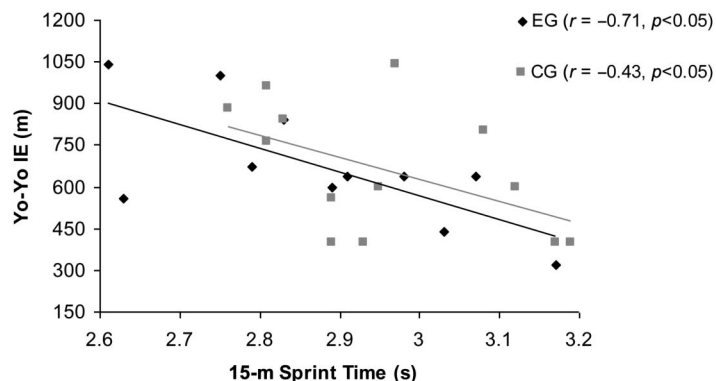
A significant increase ( $p \leq 0.05$ ) was observed in height in the CMJ in S between baseline and posttest (6.72%; ES = 0.37). Significant ( $p \leq 0.05$ ) decreases were observed in height in the CMJ in C between baseline and posttest (-10.82%; ES = 0.61). Significant differences ( $p \leq 0.05$ ) were observed after training in the magnitude of the increase between the S and C groups (Table 3).

### Yo-Yo Intermittent Endurance Test

Significant increases ( $p < 0.05$ ) were observed in the Yo-Yo IE test in S and C between baseline and posttest (49.57%, ES = 1.39; 19.67%, ES = 0.55), respectively. Significant differences ( $p \leq 0.05$ ) were observed after training in the magnitude of the increase between the S and C groups (Table 3).

### Sit and Reach Flexibility Test

A significant increase ( $p \leq 0.05$ ) was observed in the Sit and



**Figure 2.** Correlation between Yo-Yo intermittent endurance (Yo-Yo IE) test (meters) and 15-m sprint time (seconds) in the S and C groups at posttest.



Reach Flexibility test in S between baseline and posttest (7.26%; ES = 0.37). Significant ( $p \leq 0.05$ ) decreases were observed in C between baseline and posttest (-13.09%; ES = 0.94). Significant differences ( $p \leq 0.05$ ) were observed after training in the magnitude of the increase between the S and C groups (Table 3).

#### Correlations Between Variables

In the S, 15-m sprint time correlated negatively ( $p < 0.05$ ) with jump height in CMJ ( $r = -0.73$ ) (Figure 1) and with Yo-Yo IE test ( $r = -0.71$ ) (Figure 2). In the C, jump height in the CMJ correlated negatively ( $p < 0.05$ ) with the 15-m sprint time ( $r = -0.58$ ) and positively with the Sit and Reach test ( $r = 0.64$ ). No significant correlations were found between the jump height in CMJ and flexibility test (Sit and Reach) in S.

#### DISCUSSION

A novel approach in this study was to examine the effect of a 26 weeks of a combined strength and high-intensity training program in prepubertal (8–9 years) soccer players in an attempt to maximize sprint and jumping ability, intermittent endurance, and hip flexibility. Our results substantiated our hypothesis that the training intervention used (2 d·wk<sup>-1</sup>, squats, different type of jumps, and sprints exercises) improves jumping performance in a group of prepubertal soccer players. Furthermore, intermittent endurance and hip-flexibility performance were significantly enhanced in the S group after the strength and high-intensity training program. These results tend to support most of the previous published studies performed examining these types of training interventions with young soccer players (7,8,12,16,30). Although previous studies selected players older than those selected in this study and measured adaptations in shorter training interventions, generally the training strategies used confirmed that the inclusion of a more intense and nonspecific training stimulus into the normal soccer-training program produces larger jump and sprint performances than natural growth and specific soccer practice generates. Moreover, the results obtained in this study are in line with those of the researchers that justify the positive effect of a strength and high-intensity training on endurance capacity (42).

A great deal of research has focused on the development of sprint performance using a myriad of training methods, including speed training, technical sprint drills, sprinting against resistances, weight training, combined resistance and speed training, or plyometric training (30,37,39). Most related studies have selected 10- or 30-m distances for sprint testing. In this study, 15-m distance was selected because during a soccer game each sprint bout last on an average of 2–3 seconds (14). Various studies have suggested that the strength training can improve the sprint ability of young soccer players (7,16,25,26,32). For example, Chelly et al. (7) after an 8-week (2 s·wk<sup>-1</sup>) training intervention based on squat exercise (loads ranging between 70 and 95% 1RM) supplemented to the habitual soccer training observed significant

improvements in sprinting (7–12%). Additionally, positive results in sprint ability have been obtained when strength training was combined with plyometric training (11,27) or when plyometric training was applied alone (30). In a representative study because of its similarity, Meylan and Malatesta (30) observed that 8 weeks of low-intensity plyometric training (2 training interventions per week) implemented in the soccer-training program of adolescent players resulted in significantly improved 10-m sprint times (-2.1%). On the contrary, the present research suggests no significant improvement in 15-m sprint time after the 26 weeks of the S intervention. It is well known that sprint running performance is the product of stride rate and stride length with numerous components influencing this apparently simple formula. Because both elements are clearly influenced by the anthropometric characteristics (38), one of the main possible explanations for the decrement in the sprint performance observed in the S group could be the anthropometric changes observed in subjects used in this investigation. Indeed, during the experimental phase the S group increased their height significantly in respect to the C group. From the results observed in this study it is clear that control of the anthropometric state of the players in short intervals of time may be an obligatory element in further studies analyzing the effects of training intervention in young soccer players. On the other hand, performance in sprint exercise has traditionally been believed to be largely dependent on genetic factors, with only relatively small improvements occurring with training (38). Improvements in 10- and 100-m sprint times have been observed after a training intervention that incorporated some sprint-specific plyometric exercises (11). For example, jumping exercises that were nonspecific to running performance did not cause any effect on running speed (14). When exercises were specific (e.g., speed bounding) to running performance, the training program had a positive effect on running velocity (13). In this sense, a lack of specificity in the training could be another reason that may explain partly the lack of performance observed in the S group especially between baseline and T2 where it may be speculated that anthropometric changes were less significant. It is possible that a training program that incorporates greater horizontal acceleration (i.e., skipping, jumps with horizontal displacement) would result in the most beneficial effects.

Numerous studies (16,32) have measured vertical jump height as an indicator of muscle power of the lower limbs in soccer players. Several studies have shown the effectiveness of plyometric and strength training in improving vertical jump (16,30,39). On the other hand, previous studies report no improvements in the vertical jump after strength training when slow or normal contraction speed is used during training (15). In this study, significant CMJ improvements were observed between baseline and posttest. However, CMJ performance decreased in the C group. The improvements concur with those of previous studies (16,32), showing that a combined program of different modalities of strength training and power-oriented strength

training (i.e., using full-squat or parallel-squat exercise) and plyometrics can significantly increase vertical jump performance. The discrepancy between these results and results from previous studies might be attributed to several reasons: the subjects in this study were very young and not specialists in plyometric and strength training in contrast to the greater training experience and initial training status of subjects in previous research; the speed of movement rather than the resistance or load was more important and positively affects the jump performance of young soccer players; the differences in soccer players' training history (i.e., with or without systematic strength training); the competitive level or the procedures used to measure vertical jumping performance may explain these discrepancies. Specifically, some authors have shown that subjects with low levels of strength show significant improvement in vertical jump ability, regardless of the training stimulus (1,39), whereas previously strength-trained subjects may show limited improvements in vertical jump ability (17).

Several studies have shown a significant correlation between sprint time and CMJ using different strength training methods (27,34,39,43,44). In this study, a significant negative correlation ( $r = -0.77, p < 0.05$ ) was found between vertical jump height in the CMJ and 15-m sprint time. This is in accordance with previous studies in senior (45) and young (6,16) elite soccer players. For example, Wisloff et al. (46) observed in Norwegian senior elite players a significant correlation between vertical jump height and 10- and 30-m sprint time. These results confirm the relation between vertical jump height and short-duration sprint time and agree with those biomechanical analyses of sprinting showing that short-distance sprint is highly dependent on the subject's ability to generate powerful extensions of the knee extension, hip extensor, and plantar-flexors muscles. This suggests a possible transfer from the gain in the leg muscular power into the sprint performance (16).

An interesting finding in this study was that the S group increased intermittent endurance (>40%) after combined strength and high-intensity training. This agrees with the results previously described that strength and high-intensity training, in the form of dynamic exercises (i.e., squat, weighted CMJ, drop jump, and sprints), has been reported to enhance an individual's ability to rapidly develop endurance, obtain significant gains in the  $\dot{V}O_{2\max}$  and allow for greater improvement in IE (6,18,21,22,47,48). The results obtained are not compatible with the results of previous studies (9,20), which suggested an incompatibility of strength training and enhance IE. Therefore, the combined strength and high-intensity training used in this study enhanced aerobic-anaerobic and explosive performances and resulted in improved IE. Nevertheless, it is believed that the concurrent soccer training also plays a role in maintaining the IE level in both groups. Furthermore, in this study, the relationship between 15-m sprint time and Yo-Yo IE test ( $r = -0.77, p < 0.05$ ) is interesting. This is in accordance with previous studies (3,47). All these studies produce

findings that point in the same direction, independent of the age of the subjects, skill, performance, and the aim of study. It suggests that subjects get quicker and more effective recoveries encouraging new intense and effective action through strength training (3,6,18,21,23,28,47,48).

Flexibility increased significantly (7.93%) in the S and decreased significantly (-13.23%) in the C group, producing a significant difference between the groups. It seems that the group that only participated in soccer training impair flexibility development, even when proper care was taken to include stretching exercises in the training program. Recent studies concerning flexibility reported that if stretching exercises are part of the strength training program, flexibility would not be impaired and it may even increase (31,40). These studies showed similar results with other population types and strength training programs (40). In this study, stretching exercises were performed before, during, and after the strength and high-intensity training sessions. The low volume and high intensity of strength training or the combination of soccer and resistance training in our study improve flexibility. It should be noted that in the C, a significant decrease in flexibility was observed. Furthermore, our subjects were prepubertal, and the phase of their physical development might also interact with flexibility levels. More emphasis was probably required on specific stretching exercises and further research is needed to determine which factors might impede flexibility or not when prepubertal soccer players lift weights and follow strength and power-oriented training programs.

In this study, the injury rate was only 3 muscular or ligament injuries. The rates are relatively low, considering the training mode, (i.e., low-impact plyometric exercises, squat, and sprint exercises). Higher injury frequencies have even been encountered in football and in other sports (10,29). Any interruptions in training because of musculoskeletal symptoms and injuries were short, suggesting that the disorders and injuries were not serious. The cornerstones of the training were throughout the intervention sufficient warming-up before training, muscle stretching after training, not too fast progressing intensity, variation in training sessions, and finally, no competitive elements were included in the training program. The relatively low incidence of training induced injuries and the unchanged or decreased level of musculoskeletal symptoms during the training indicate the feasibility of the program.

In summary, the present data clearly demonstrated that adding strength and high-intensity training in previously moderately trained prepubertal soccer players appear to be a good stimulus for improving jumping ability, intermittent endurance, and flexibility performance. Moreover, for very young soccer players who do not have prior experience with strength and high-intensity training, a general adaptation phase must be scheduled to ensure proper movement technique and safety. As a result, the applicability of strength and high-intensity training together with regular soccer



training could be performed during the season with no concomitant interference on endurance performance.

## PRACTICAL APPLICATIONS

We have clearly demonstrated that prepubertal soccer players can enhance muscle strength, power, and endurance by undertaking a 26-week in-season program of combined strength and high-intensity-oriented training involving exercises for lower body (squat, full-squat, loaded and unloaded jump, and sprint exercises). Moreover, there is no apparent interference between the development of strength and power and endurance and flexibility performance. This study also emphasizes the importance of high levels of strength and power to resolve high-intensity game actions. Such benefits can be realized from only 2 short strength and high-intensity training sessions per week in-season. The performance improvements shown in this study are of great interest for soccer coaches and are directly applicable to prepubertal soccer players, because the performance of this sport relies greatly on the specific on-field vertical jump, maximal sprint, and agility abilities that were enhanced by the lower-body strength and high-intensity oriented training regimen. Previous authors have found a similar benefit of strength and power training in others and this sport, but this is the first study to our knowledge involving prepubertal soccer players. It is recommended that soccer coaches implement in-season strength and high-intensity training to enhance the performance of their players. The outcomes may help coaches and sport scientists formulate better guidelines and recommendations for athlete assessment and selection, training prescription and monitoring and preparation for competition. Similar studies using larger group numbers plus more extensive preparatory strength programs may produce results indicating the superiority of  $\geq 1$  of these training modes.

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